Data Structure Assignment

Question for 3rd A & B Sec 2025

Q. 1. Define Data structure, With a neat diagram Explain the classification of Data structure with Example.

Q 2. Explain the memory allocation. Explain Dynamic memory allocation functions supported by 'C' with syntax and Example.

Q 3. Explain structure and Union with an Example. (Differentiate both) list out differentiate structure and union.

Q 4. Explain any four String handling function supported by 'C' with syntax and Example.

Q 5. Consider the pattern P = abababa. Construct the table and the corresponding labeled directed graph used in the fast or second pattern matching algorithm. Trace it for the input text T = abcabababa.

Or

5. What do you mean by pattern matching? outline the Knuth morris pratt (KMP) algorithm and illustrate it to find the occurrences of the following pattern.

P : ABCDABD

S : ABC ABCDAB ABCDABCDABDB.

Q 6. ~~Write function in C~~

Q 6. Write a program in C to implement Push, Pop and display operation for stack using arrays.

Q 7. Write function in C for the following operations without using built-in functions.

i) Compare two strings.
ii) Concatenate two strings.
iii) Reverse a string.

Q. 8. Explain the representation of linear Arrays in memory. Also, consider the linear arrays AAA(5:50) and BBB(-5:10).

i) Find the number of elements in each array.
ii) Suppose Base(AAA)=300 Base(BBB)=500 and 4 words per memory cell for AAA, 2 words per memory cell for BBB, find the address of AAA[15], AAA[55], BBB[8], and BBB[0].

Q 9. Define a stack. Explain the different operations that can be performed on stacks with.

Q 9. Convert the following infix expression into postfix Expression using Stack.

A + (B * C - (D / E ^ F) * G) * H.

Q 10. Write an algorithm to evaluate a postfix expression. Trace the algorithm for the following expression showing the stack contents

6 5 1 - 4 * 2 3 ^ / +

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11. Illustrate the same for the given postfix expression.

A B C - D * + E $ F + and assume

A = 6, B = 3, C = 2, D = 5, E = 1, and F = 7

12. Develop a C program to implement - insertion, deletion and display operations on Linear queue.

13. Write the C function to add two polynomials. Show the linked representation of the below two polynomials and their addition using a circular Singly linked list.

P_1 : $5x^3 + 4x^2 + 7x + 3$
P_2 : $6x^2 + 5$

output: add the above two polynomials and represent them using the linked list.

14. Develope a C recursive program for factorial of N number.

16. Develop C functions to implement insertion, deletion and display operations of a Circular queue.

17. Explain link list. Write an algorithm to demonstrate single linked list and doubly linked list.

Explain queue. Explain Circular queue and Give an function per Circular queue.